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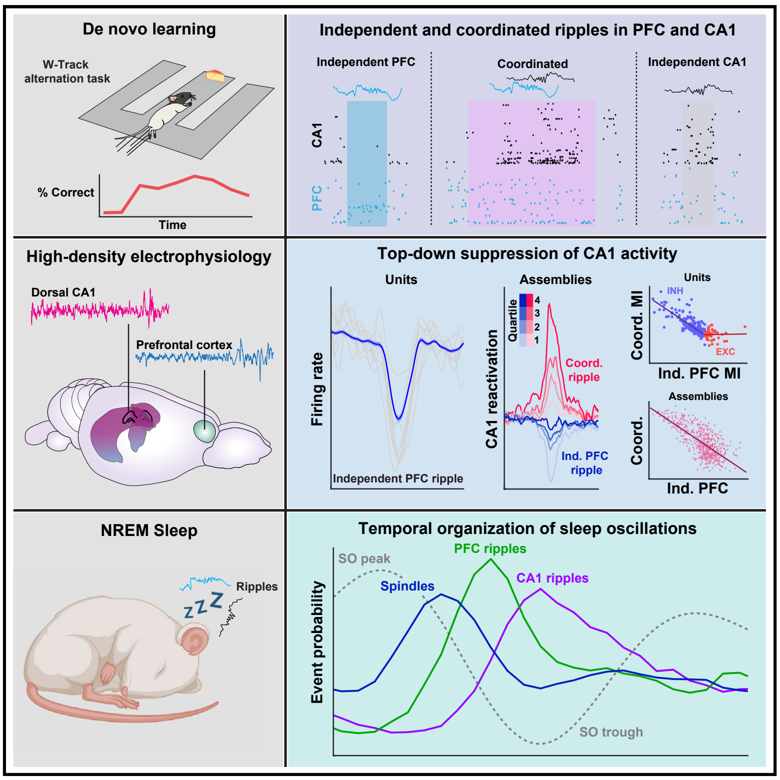
Prefrontal cortical ripples mediate top-down suppression of hippocampal reactivation during sleep memory consolidation

Graphical abstract



Highlights

- Independent and coordinated ripple events across CA1 and PFC in NREM sleep
- Suppression of hippocampal reactivation during independent PFC ripples
- Hippocampal suppression is inversely related to coordinated ripple reactivation
- Serial organization of slow oscillations, spindles, and cross-regional ripples

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In brief

Shin and Jadhav report that independent prefrontal cortex (PFC) ripples predominantly suppress CA1 activity in NREM sleep when temporally dissociated from hippocampal sharp-wave ripples (SWRs). These suppressed CA1 representations are strongly reactivated during coordinated CA1-PFC ripples, which are serially organized with slow oscillations and spindles.

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Prefrontal cortical ripples mediate top-down suppression of hippocampal reactivation during sleep memory consolidation

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SUMMARY

Consolidation of initially encoded hippocampal representations in the neocortex through reactivation is crucial for long-term memory formation and is facilitated by the coordination of hippocampal sharp-wave ripples (SWRs) with cortical slow and spindle oscillations during non-REM sleep. Recent evidence suggests that high-frequency cortical ripples can also coordinate with hippocampal SWRs in support of consolidation; however, the contribution of cortical ripples to reactivation remains unclear. We used high-density, continuous recordings in the hippocampus (area CA1) and prefrontal cortex (PFC) over the course of spatial learning and show that independent PFC ripples dissociated from SWRs are prevalent in NREM sleep and predominantly suppress hippocampal activity. PFC ripples paradoxically mediate top-down suppression of hippocampal reactivation rather than coordination, and this suppression is stronger for assemblies that are reactivated during coordinated CA1-PFC ripples for consolidation of recent experiences. Further, we show non-canonical, serial coordination of independent cortical ripples with slow and spindle oscillations, which are known signatures of memory consolidation. These results establish a role for prefrontal cortical ripples in top-down regulation of behaviorally relevant hippocampal representations during consolidation.

INTRODUCTION

Systems memory consolidation is the process through which newly formed hippocampal memories are integrated in cortical networks for long-term storage, and models of memory consolidation, such as the standard systems theory of consolidation and multiple trace theory, posit a predominantly unidirectional interaction between the hippocampus and cortical regions for consolidation.[1–3] This consolidation process is facilitated through the coordination of patterns of activity that span a range of timescales and oscillatory frequencies and has been extensively studied in the context of the hippocampal-prefrontal cortical network. The most prominent of these activity patterns are hippocampal sharp-wave ripples (SWRs), a phenomenon critical for memory consolidation.[4] These high-frequency bursts during non-REM (non-rapid eye movement [NREM]) sleep reactivate hippocampal representations of previous experiences, for example, in the form of sequential reactivation of spatial representations, engage the prefrontal cortex (PFC), and are associated with brain-wide changes in activity hypothesized to support the integration of mnemonic representations in distributed cortical circuits.[5–8] Recently, high-frequency ripple bursts have been reported in the neocortex, including in the PFC, which synchronize local cortical activity and can potentially be coupled to hippocampal SWRs.[9,10] This coupling of ripples can serve to support consolidation through coordinated reactivation; however, whether and how PFC ripples reciprocally engage hippocampal activity—and especially hippocampal reactivation—in support of consolidation is unclear.

RESULTS

Independent and coordinated ripple events across CA1 and PFC

Here, we simultaneously recorded populations of CA1 (n = 3,316) and PFC (n = 2,474) neurons (Figures S1A and S1B show example histology) from animals (n = 8 rats) in NREM sleep (mean = 13.24 ± 0.83 min per epoch; Figure S1C; Table S1) throughout the course of learning a spatial alternation task that requires hippocampal-prefrontal interactions.[11,12] Activity was monitored across multiple run and sleep sessions during learning. We detected cortical ripples in the PFC as high-frequency events (150–250 Hz), as previously described.[9,10] Because these events could occur either in an isolated manner or in temporal proximity to hippocampal ripples, we classified them as either temporally coupled to hippocampal SWRs (i.e., coordinated ripples) or independent (Figures 1A–1D). We thus separated ripples, based on the interregional overlap of these events, as independent PFC ripples (71.2% of events), independent CA1 SWRs (termed independent SWRs; 78.7% of events),

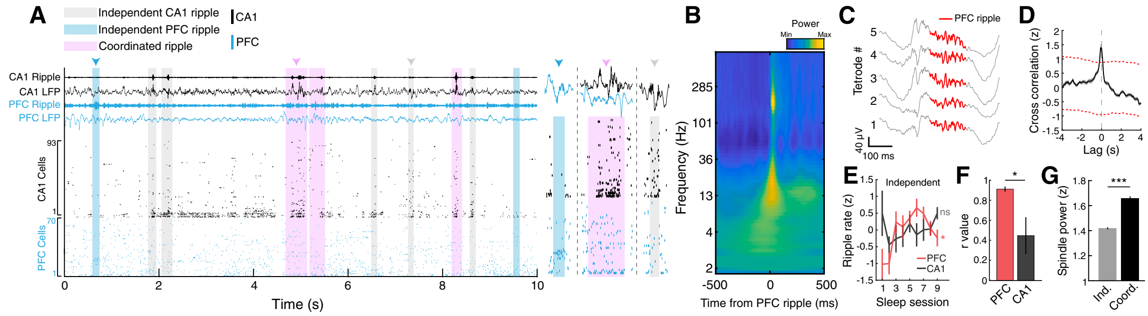

Figure 1. Independent and coordinated ripple events in CA1 and PFC

(A) Example CA1 and PFC raster plot illustrating the occurrence of independent CA1 and PFC ripples and coordinated events. Events indicated by the arrowheads are expanded on the right to show local field potential (LFP) and single unit activity.

(B) PFC-ripple-triggered spectrogram combined across all animals (n = 8 rats) and epochs (n = 72 epochs; n = 37,335 ripples). Note the associated spindle band activity (12–16 Hz) during these events.

(C) Example PFC ripple event shown across 5 tetrodes. Detected ripple event is highlighted in red.

(D) Cross-correlation between all ripple events in CA1 and PFC. Red dashed lines indicate the 95% confidence intervals for jittered data.

(E) Independent ripple rate over the 9 sleep epochs (left, PFC: F = 2.44, *p = 0.03; CA1: F = 1.18, p = 0.34, main effect of time, repeated measures ANOVA).

(F) The correlations between overall and coordinated ripple rate over time for CA1 and PFC events suggest PFC-driven coupling (PFC, r = 0.91 ± 0.03; CA1, r = 0.45 ± 0.18, *p = 0.01).

(G) Peak spindle power during independent vs. coordinated PFC ripple events (independent = 1.42 ± 0.01, coordinated = 1.66 ± 0.01, ***$p = 2.11 \times 10^{-66}$).

See also Figures S1 and S2 and Table S1.

and coordinated CA1-PFC ripples (CA1, 21.3% of events; PFC, 28.8% of events; Figure S1). Intriguingly, independent PFC ripples and independent CA1 SWR events were more prevalent than coupled CA1-PFC ripple events in NREM sleep (Figure S1E; independent PFC ripple rate: 0.46 ± 0.02 Hz; independent CA1 ripple rate: 0.67 ± 0.02 Hz; coordinated ripple rate: 0.16 ± 0.01 Hz). We hypothesized that these two kinds of events, independent and coordinated, may subserve different mnemonic processes within the hippocampal-prefrontal network and, indeed, found key differences. First, coordinated events were longer in duration and had higher cell participation (Figures S2A and S2C), consistent with previous reports that long-duration events may be important for performing behavioral tasks with high memory demand.[13] In contrast to CA1 SWRs, the rate of independent PFC ripples increased over time from pre-exploration sleep periods in animals that learned the task and was correlated with task performance in the subsequent behavioral session (Figures 1E, S1F, and S1J–S1M). The rate of coordinated CA1-PFC ripples also increased with learning and was correlated with the increase in overall PFC ripple rate, suggesting that the PFC plays a role in driving the increase in coupling during later sleep epochs (Figures 1F and S1I). Finally, coordinated CA1-PFC ripples were also associated with higher spindle power (Figure 1G), indicating enhanced cross-frequency coupling between the hippocampus and PFC during coordination, which is critical for memory consolidation.[14–16]

PFC ripples predominantly suppress hippocampal activity during NREM sleep

Previous reports have shown that hippocampal SWRs are associated with bidirectional modulation of PFC neurons and PFC reactivation of recent experiences.[12,17–19] Here, we examined the effect of independent PFC ripples on CA1 neuronal activity and the influence of cortical ripples on reactivation. Interestingly, a large majority of CA1 cells, both putative pyramidal cells (inhibited neuron [INH], n = 365; excited neuron [EXC], n = 140) and interneurons (INH, n = 105; EXC, n = 23; classification shown in Figures S2F–S2M), were suppressed during independent PFC ripples, while PFC activity was enhanced by the local ripples (INH, n = 99; EXC, n = 927; Figures 2A and 2B). In addition, suppression in CA1 during PFC ripples peaked prior to PFC excitation (Figure 2C), indicating attenuation of CA1 activity through a cortical-hippocampal-cortical loop that may enhance information transfer within the neocortex.[20,21] Additionally, we observed a relationship between hippocampal suppression and the frequency of independent PFC ripple events, potentially indicative of a gradient of hippocampal-cortical engagement, as previously suggested[22] (Figure 2D). This neuronal activity suppression was dependent on the ripples examined (Figures 2A and S3), either local ripples in CA1/PFC or coordinated ripples in CA1-PFC (Figures 3D and S3A), with coordinated ripples predominantly driving excitation in both regions in contrast to independent cortical ripples, indicating that these temporally distinct events differ in their contribution to mnemonic processes.

PFC ripple modulation reveals distinct populations of CA1 neurons

Further examination of the inhibited and excited CA1 pyramidal neurons during run sessions revealed representational differences (Figures 3A and 3B). INH cells had higher theta firing rates, had wider and more numerous fields, and had greater stability between sessions, whereas EXC cells were more strongly locked to the hippocampal theta oscillation (Figures 3B and S3G). During NREM sleep, EXC and INH cells had similar baseline firing rates but had distinct intra-SWR firing dynamics. EXC cells were generally active earlier in SWR events (spike latency plot in Figure 3C) and INH cells were more strongly SWR excited (firing rate gain and burst probability plots in Figure 3C). Importantly, the degree of CA1 modulation by PFC ripples was also predictive of intra-CA1-SWR dynamics (Figure 3C, right) and

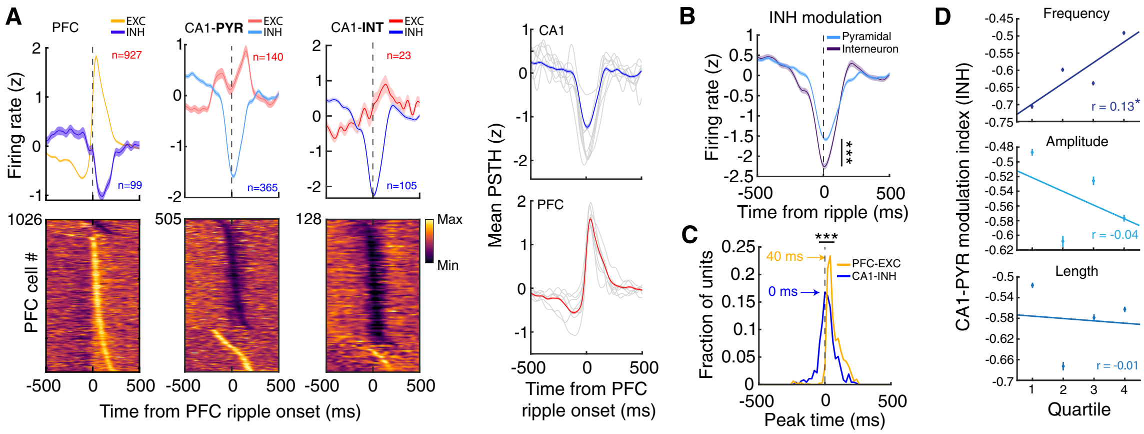

Figure 2. Predominant suppression of CA1 activity during independent PFC ripples

(A) Independent PFC-ripple-triggered modulation of PFC neurons (left), CA1 pyramidal neurons (middle), and CA1 interneurons (right). (Right) Overall modulation in each area showing neurons excited or suppressed during ripples, with excitation predominantly in the PFC and suppression predominantly in CA1 (pyramidal and interneurons). Gray lines indicate individual animals.

(B) The degree of suppression was higher in INH interneurons as compared with pyramidal cells (±200 around ripple onset; ***$p = 1.37 \times 10^{-9}$).

(C) Timing of peak suppression for CA1 INH cells (pyramidal and interneurons) and PFC EXC cells (comparison of peak timing distributions, ***$p = 4.90 \times 10^{-46}$).

(D) Relationship between CA1 INH modulation and PFC ripple frequency, amplitude, and length (MI vs. frequency, r = 0.13, *$p = 0.018$; MI vs. amplitude, r = −0.04, $p = 0.41$; MI vs. length, r = −0.01, $p = 0.90$).

See also Figures S2 and S3.

was negatively correlated with modulation during coordinated CA1-PFC ripples (Figures 3D and 3E), indicating a role for PFC ripples in influencing the content of hippocampal activity during coordinated ripples. In particular, the suppression of those CA1 INH cells by PFC ripples that are strongly recruited during coordinated CA1-PFC ripples suggests that independent PFC ripples primarily suppress hippocampal reactivation of recent experiences during spatial learning, contrary to current models of systems consolidation. Further, a similar relationship of suppression was also found in the population of putative CA1 interneurons, demonstrating a broad top-down influence of PFC ripples on CA1 activity (Figures S3B–S3D). Thus, PFC ripples delineate transient intervals where hippocampal activity is largely suppressed, potentially modifying circuit dynamics for effective information processing or transfer.

PFC ripples mediate top-down suppression of CA1 reactivation

We then examined population activity in CA1 and PFC during these ripple events using assembly analysis to explicitly look at reactivation during ripples (Figures 4A and 4C). Corroborating the single unit results, assemblies within a region (CA1 $n = 585$, PFC $n = 320$) were robustly reactivated during their respective local ripple events (Figures 4B, 4D, S4B, and S4C) and were strongly co-active during coordinated CA1-PFC ripple events, as expected for a memory consolidation process (Figures S4D and S4G). CA1 EXC cells had higher weights and contribution to reactivation strength than INH cells (Figures 4D and S4H), possibly owing to the stronger theta phase locking within this population during run sessions (Figures 3B and S3G). Assemblies in CA1 and PFC exhibited gradients of reactivation strengths across events (Figures S4I and S4J) and, similar to activity suppression, the degree of reactivation strength in CA1 was negatively correlated across independent PFC ripples and coordinated CA1 SWRs (Figure 4E), confirming that assemblies that are strongly reactivated during coordinated CA1-PFC ripples are highly suppressed by independent PFC ripples. This CA1 suppression was also strongest during the later sleep sessions when the PFC ripple rate peaked (Figures 1E and S4E). Interestingly, the degree of suppression of CA1 assemblies during independent PFC ripples was related to assembly reinstatement in subsequent behavioral sessions (Figure 4G), suggesting a role for this top-down suppression in modifying subsequent CA1 coactivity patterns expressed during behavior. In addition, both reactivation strength and within-area coactivity in PFC and CA1 differed across ripple events, and population activity in PFC and CA1 was able to distinguish between independent and coordinated ripples in each respective area, confirming their distinct activity profiles and reactivation signatures (Figures 4F, 4H, and S4F). We observed similar relationships for joint CA1-PFC reactivation strength. Joint CA1-PFC coactivity differed between independent and coordinated SWRs (Figures S5A–S5D) and also showed a negative relationship between reactivation strength during coordinated ripples vs. independent PFC ripples (Figures 4I–4K). These results indicate that independent and coordinated ripples underlie different modes of information exchange between CA1 and PFC and, further, that the stronger the assembly reactivation in CA1 during coordinated CA1-PFC ripples, the stronger the suppression during PFC ripples.

The content of CA1 SWRs is dependent on cross-regional, high-frequency coordination

Because it has been previously demonstrated that hippocampal reactivation and replay are separable components of the dynamic hippocampal code,[23] we examined how association with PFC ripples may impact CA1 replay. We found that replay events associated with coordinated ripples were less sequential in structure, having higher normalized jump distances and lower

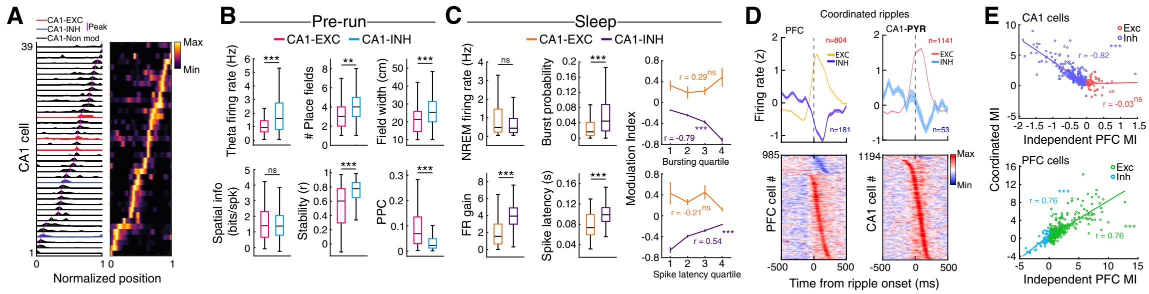

Figure 3. Properties of PFC-ripple-modulated CA1 pyramidal neurons

(A) Example firing maps for CA1 EXC, INH, and non-modulated cells during one trajectory.

(B) Differences in spatial properties for CA1 EXC and INH pyramidal cells during the run session preceding the sleep epoch in which the cells were modulated (theta firing rate, EXC = 1.20 ± 0.10 Hz, INH = 2.05 ± 0.10 Hz, $^{***}p = 2.08 \times 10^{-6}$; number of fields, EXC = 3.3 ± 0.17, INH = 3.99 ± 0.11, $^{**}p = 0.002$; field width, EXC = 21.67 ± 0.91 cm, INH = 26.56 ± 0.61 cm, $^{***}p = 1.1 \times 10^{-4}$; spatial information (bits/spike), EXC = 1.52 ± 0.12 bits, INH = 1.49 ± 0.06 bits, $p = 0.83$; stability, EXC = 0.54 ± 0.03, INH = 0.72 ± 0.01, $^{***}p = 9.01 \times 10^{-7}$; pairwise phase consistency, EXC = 0.10 ± 0.01, INH = 0.05 ± 0.004, $^{***}p = 3.31 \times 10^{-10}$).

(C) (Left) Mean NREM firing rate (FR) and intra-SWR firing dynamics for CA1 EXC and INH cells during coordinated ripples (NREM firing rate, EXC = 1.07 ± 0.11 Hz, INH = 0.81 ± 0.05 Hz, $p = 0.18$; FR gain, EXC = 2.02 ± 0.16, INH = 4.07 ± 0.10, $^{***}p = 2.88 \times 10^{-27}$; burst probability, EXC = 0.031 ± 0.006, INH = 0.075 ± 0.005, $^{***}p = 6.38 \times 10^{-5}$; spike latency, EXC = 0.080 ± 0.006, INH = 0.099 ± 0.001, $^{***}p = 1.61 \times 10^{-5}$). (Right) Relationship between SWR bursting incidence (top, EXC: r = 0.29, $p = 0.08$, INH: r = −0.79, $^{***}p = 5.20 \times 10^{-69}$) or spike latency for CA1 EXC and INH cells (bottom, EXC: r = −0.21, $p = 0.21$, INH: r = 0.54, $^{***}p = 9.51 \times 10^{-25}$) with modulation index (MI) during independent PFC ripple events.

(D) Modulation of PFC and CA1 neurons during coordinated ripples, aligned to ripple onset in the opposing-area.

(E) Relationship between MI during coordinated ripples and independent PFC ripples (top, CA1, EXC: r = 0.03, $p = 0.82$; INH: r = −0.82, $^{***}p = 6.84 \times 10^{-80}$; bottom, PFC, EXC: r = 0.76, $^{***}p = 1.34 \times 10^{-92}$; INH: r = 0.76, $^{***}p = 1.51 \times 10^{-10}$). Note the negative relationship for CA1 INH cells, indicating that the stronger the recruitment during coordinated ripples, the more inhibited the cells are during independent PFC ripples.

See also Figure S3.

weighted correlations than independent-SWR-associated replay (Figures 5A, 5B, and S5E–S5G). Sequential unit spiking was also more consistent, as measured by rank-order correlation,[24] and had higher dimensionality during independent SWRs, further highlighting the influence of the PFC on SWR content (Figures 5C and 5D). Coordinated SWR replay exhibited a larger degree of sequence degradation when place-cell templates are shuffled as compared with independent SWR and awake replay (Figure 5E), possibly due to a decrease in overall reactivation strength and coactivity during these events (Figures 4F and S4F). Furthermore, PFC-ripple-modulated CA1 cells had a higher per-cell contribution (PCC) to replay than non-modulated cells (Figure 5F, top). Although both modulated and non-modulated cells contributed more strongly to coordinated SWR replay, which is expected due to the greater sequence degradation observed in these events, modulated cells had higher PCC differences (Figure 5F, bottom). This overall higher contribution of modulated cells to coordinated SWR replay, and the differences observed between independent and coordinated events, further suggests a role for PFC in shaping the content of hippocampal reactivation and replay during sleep.

Sleep oscillations exhibit distinct sequential coupling for cortical ripples

In addition to the recent observation of temporal coupling of hippocampal-cortical ripples contributing to memory consolidation, the coordination of hippocampal SWRs with cortical slow oscillations (SOs) and spindles has been established to be important for consolidation.[14,25,26] To this end, we investigated the incidence of ripple events in the context of these oscillations that span multiple timescales. Independent and coordinated CA1 SWRs exhibited differential coupling with spindles and SOs, with only coordinated CA1-PFC ripples associated with cortical spindles and not independent CA1 SWRs, and different timing profiles of the two types of ripple events with SOs (Figures 6A, S6A, and S6B). Population activity in CA1 and PFC also had unique responses surrounding independent vs. coordinated events (Figures S6C and S6D). Additionally, CA1 replay events, specifically during coordinated CA1-PFC ripples, were enriched during "up-states" of the SO (Figures 6B, 6C, and S6F). These results, along with the finding that coordinated CA1-PFC ripples were associated with a higher cortical spindle power (Figure 1G), led us to hypothesize distinct temporal organization of independent and coordinated ripple events with SOs and spindles.

We observed an overall higher prevalence of coordinated events where PFC ripples precede CA1 SWRs (Figure S6J), which is consistent with PFC ripples potentially driving ripple coordination (Figure 1F), and therefore focused on these events for the subsequent analyses to investigate sequential coupling of events in CA1 and PFC. We quantified the onset of PFC spindles, ripples, and CA1 SWRs relative to the trough (up-state) of SOs and calculated the fold change probability of each event around SOs. We found a sequential organization of these oscillatory events that suggests a thalamocortical-hippocampal directionality, with spindles preceding coordinated PFC-CA1 ripple events (Figures 6D and 6E) as well as independent PFC ripples (Figure S6H). This temporal relationship with spindles was not present for independent CA1 SWRs (Figure S6H) and differed for coordinated events where CA1 SWRs preceded PFC ripples (Figure S6I). The embedding of PFC ripple events between spindles and CA1 SWRs (Figures 6E and 6F) would allow for the coordination of PFC and CA1 assembly reactivation. Indeed, PFC and CA1 reactivation was temporally organized

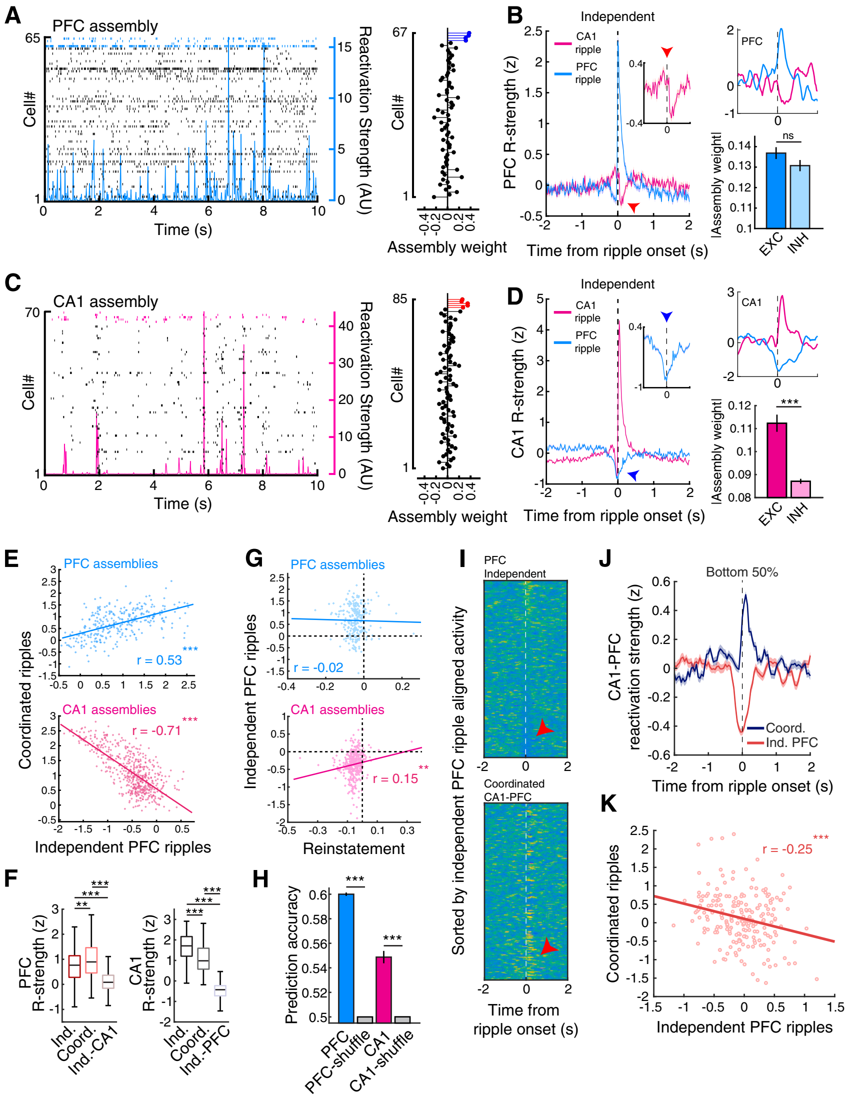

Figure 4. Assembly reactivation during independent and coordinated ripple events

(A) (Left) Example PFC raster plot during NREM sleep with the reactivation strength of an assembly overlaid. (Right) Neuron weights for the plotted assembly on the left, with member cells highlighted in blue.

(B) (Left) Independent PFC- and CA1-ripple-triggered PFC reactivation strength with inset highlighting cross-regional modulation. (Right) Example PFC assembly aligned to CA1 and PFC ripple events (top) and the absolute weights of EXC and INH PFC cells (bottom; modulation during independent CA1 SWRs, EXC = 0.137 ± 0.003, INH = 0.131 ± 0.003, $p = 0.37$).

(C and D) Same as in (A) and (B) but for CA1 (modulation during independent PFC ripples, EXC = 0.112 ± 0.004, INH = 0.087 ± 0.001, $^{***}p = 8.04 \times 10^{-7}$).

(E) Relationship between Z scored assembly reactivation strength during independent PFC ripples and coordinated ripples (PFC: r = 0.53, $^{***}p = 1.28 \times 10^{-23}$; CA1: r = −0.71, $^{***}p = 4.25 \times 10^{-87}$). Note that PFC reactivation strength is correlated across events, whereas CA1 assemblies that are highly suppressed during independent PFC ripples are strongly reactivated during coordinated ripples.

(F) Reactivation strength during within-area independent and coordinated ripples and during opposing-area independent events (PFC: independent PFC = 0.71 ± 0.03, coordinated PFC = 0.96 ± 0.04, independent CA1 = 0.13 ± 0.02, $^{**}p = 0.0023$, $^{***}p = 3.21 \times 10^{-33}$, $^{***}p = 2.12 \times 10^{-53}$ for independent PFC vs. coordinated PFC, independent PFC vs. independent CA1, and coordinated PFC vs. independent CA1, respectively; CA1: independent CA1 = 1.67 ± 0.03, coordinated CA1 = 1.11 ± 0.03, independent PFC = −0.51 ± 0.02, $^{***}p = 2.44 \times 10^{-18}$, $^{***}p = 7.45 \times 10^{-239}$, $^{***}p = 1.14 \times 10^{-128}$ for independent CA1 vs. coordinated CA1, independent CA1 vs. independent PFC, and coordinated CA1 vs. independent PFC, respectively; Kruskal-Wallis test, Bonferroni corrected for multiple comparison).

(G) Relationship between independent PFC ripple reactivation and reinstatement of assemblies in the subsequent run epoch (PFC: r = −0.002, $p = 0.76$, $n = 257$; CA1: r = 0.15, $^{**}p = 0.0031$, $n = 382$).

(H) Prediction of intra-area ripple type (independent or coordinated) using spike count data (PFC: data = 0.6 ± 0.001, shuffle = 0.5 ± 4.64×10^{-5}, $^{***}p = 3.39 \times 10^{-50}$; CA1: data = 0.55 ± 0.005, shuffle = 0.50 ± 1.37×10^{-4}, $^{***}p = 1.28 \times 10^{-26}$).

(I) All CA1-PFC joint assemblies sorted by independent-PFC-ripple-aligned reactivation strength. Assemblies must have at least one member cell in each area to be included in analysis. Note that the high degree of CA1-PFC assembly suppression during independent PFC ripples is associated with strong reactivation during CA1-PFC coordinated ripples (red arrowheads).

(J) The bottom 50% of assemblies sorted by independent-PFC-ripple-aligned reactivation strength.

(K) CA1-PFC reactivation strengths during independent PFC ripples were negatively correlated with reactivation during CA1-PFC coordinated events, similar to what is seen in CA1 assemblies (r = −0.25, $^{***}p = 3.69 \times 10^{-4}$).

See also Figures S4 and S5.

surrounding SO troughs in a manner that is consistent with cortico-hippocampal directionality (Figure 6G). In addition, the strength of the up-state-aligned reactivation in CA1 assemblies was related to the degree of suppression during independent PFC ripples (Figure 6H), suggesting a role of PFC ripples in prioritizing certain CA1 memory traces for long-term storage or modification. Additionally, differences in coordinated events (PFC leading vs. lagging), such as the timing and content of CA1 and PFC reactivation (Figures S6K and S6L), suggests a bidirectional coordinated ripple-mediated interaction with distinct roles in consolidation.

DISCUSSION

Memory consolidation is a dynamic and multidimensional process that is dependent on the interaction between multiple brain systems, and previous studies have shown that enhanced cross-frequency coupling between the hippocampus and PFC enhances performance on memory-related tasks.[14,27] Notably, the suppression of CA1 reactivation by independent PFC ripples is unexpected based on current models of systems consolidation,[1–3] and how activity suppression may contribute to consolidation in the context of interregional communication is unclear.

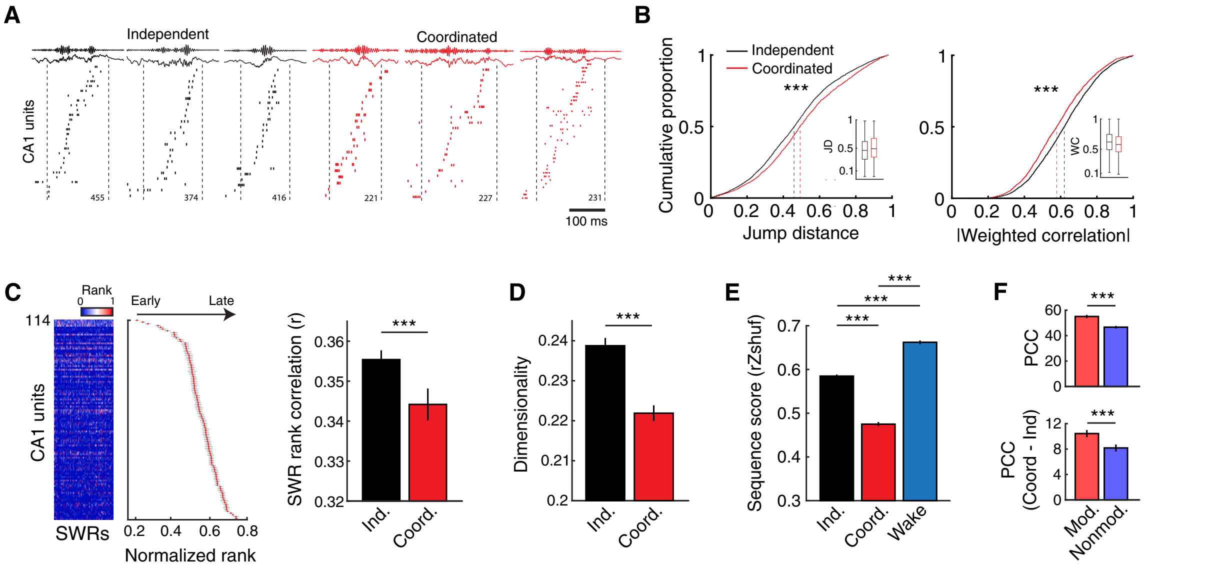

Figure 5. Differences in content between independent and coordinated SWR replay events

(A) Example CA1 rasters during independent (black) and coordinated (red) SWR events. Only significant replay events are shown.

(B) Jump distance and absolute weighted correlation for significant CA1 replay events decoded during independent CA1 SWRs and coordinated ripples (jump distance: independent = 0.468 ± 0.003, coordinated = 0.505 ± 0.005, ***$p = 2.73 \times 10^{-11}$; weighted correlation: independent = 0.620 ± 0.002, coordinated = 0.585 ± 0.003, ***$p = 1.10 \times 10^{-17}$).

(C) Rank-order correlation analysis. Example rank-order matrix and template from an example sleep epoch (left). Templates were generated separately for independent and coordinated ripple events. The order of unit spiking (both pyramidal cells and interneurons included) was more consistent for independent SWR events as compared with coordinated SWRs (independent = 0.356 ± 0.002, coordinated = 0.344 ± 0.004, ***$p = 3.34 \times 10^{-5}$, Wilcoxon rank sum).

(D) Population dimensionality was higher during independent SWRs as compared with coordinated SWRs (independent = 0.239 ± 0.002, coordinated = 0.222 ± 0.002, ***$p = 4.37 \times 10^{-15}$, Wilcoxon rank sum).

(E) Sequence degradation (rZ_{shuf}) of independent, coordinated, and awake replay events after shuffling individual cells' linearized firing fields (independent = 0.585 ± 0.004, coordinated = 0.475 ± 0.005, wake = 0.662 ± 0.004, ***$p = 7.34 \times 10^{-79}$, ***$p = 5.70 \times 10^{-42}$, ***$p = 2.40 \times 10^{-169}$ for independent vs. coordinated, independent vs. wake, and coordinated vs. wake, respectively; Kruskal-Wallis test, Bonferroni corrected for multiple comparison).

(F) Per-cell contribution (PCC) to all significant replay events for CA1 modulated vs. non-modulated cells (top, modulated = 55.04 ± 1.28, non-modulated = 46.65 ± 1.02, ***$p = 2.39 \times 10^{-7}$) and the difference in PCC (coordinated minus independent) to independent and coordinated events (bottom, modulated = 10.44 ± 0.57, non-modulated = 8.17 ± 0.55, ***$p = 9.88 \times 10^{-4}$).

See also Figure S5.

This novel, suppressive signature may have implications for understanding the time course over which circuit modifications are implemented to support memory consolidation. The relatively rapid rise and fall of PFC ripple occurrence over the time period of sleep indicates that the strongest effects of these PFC-ripple-mediated modulatory processes are transient and restricted to initial novel learning. This neurophysiological phenomenon is situated in an ideal temporal window to index or tag certain hippocampal memory traces based on previously stored, generalized representations in distributed cortical circuits.[28,29] Thus, the robust, broad suppression observed here may be an important early-onset, top-down process that converges on the hippocampus concurrently with the entorhinal cortex information streams.[30] The present results are in line with recent findings that suggest the presence of parallel processing circuits constituting both bottom-up and top-down information streams that coincide for effective memory formation and retrieval.[31] Indeed, initial learning is associated with the activation and rapid stabilization of PFC representations,[12,32] potentially recruiting a subset of experience-salient ensembles to dynamically influence ongoing learning-related plasticity in the hippocampus and the content of what information to consolidate through the coordination of cortical up-states, thalamocortical spindles, and ripples during NREM sleep.

Our finding that independent and coordinated events across the hippocampal-prefrontal circuit elicit differential responses suggests separate roles in systems memory consolidation. Although coordinated ripple events across hippocampal-neocortical circuits are thought to support memory consolidation through the coordinated reactivation of behaviorally relevant representations,[9,19] the roles of independent events in CA1 and PFC are unclear. One possibility is that independent CA1 SWRs (independent meaning uncoupled from PFC ripple activity, specifically) preferentially engage other brain areas to support different aspects of consolidation, depending on the nature of the experience. Indeed, there are widespread, bidirectional changes in activity surrounding SWRs[5,33] that could reflect distinct processes if examined on an event-by-event basis. Future studies investigating multiple regions of interest simultaneously will allow for the discretization of events spanning a wide range of brain areas and may reveal a multiplex hierarchy within which specific components can be selectively engaged depending on mnemonic demands.[34] Notably, the ability to distinguish SWR events based on long-range interregional

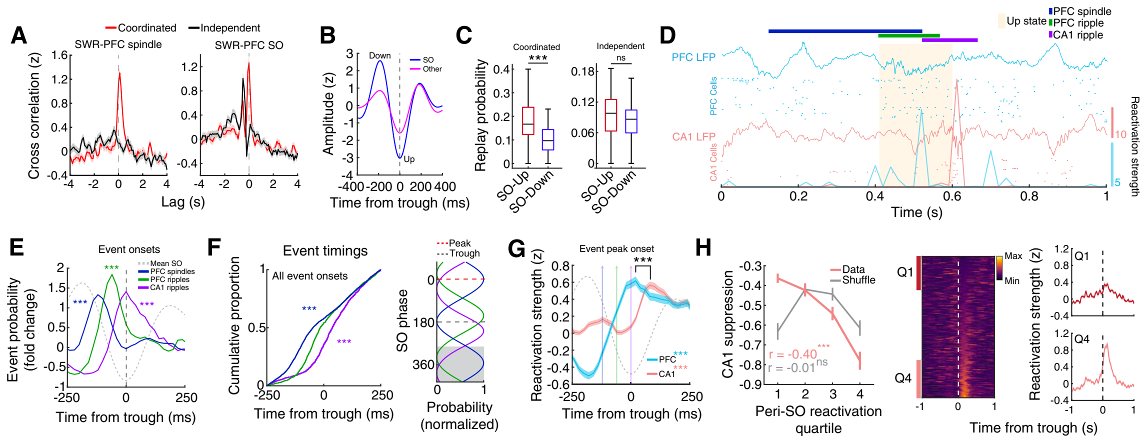

Figure 6. Serially coupled spindles, ripples, and slow oscillations

(A) Cross-correlation between independent or coordinated CA1 SWRs with PFC spindles and SOs (trough time). Note that only coordinated CA1-PFC ripples tend to be associated with spindle oscillations. Furthermore, the timing of SWRs relative to SOs is different between these events, with coordinated ripples occurring primarily at the trough and independent SWRs preceding SO troughs.

(B) Average waveforms for SOs and other, lower-amplitude slow waves extracted from PFC LFPs.

(C) Probability of coordinated- and independent-ripple-associated CA1 replay during up and down states (coordinated: up = 0.19 ± 0.02, down = 0.11 ± 0.01, ***$p = 2.30 \times 10^{-4}$; independent: up = 0.09 ± 0.01, down = 0.08 ± 0.01, $p = 0.18$).

(D) Raster showing the reactivation strengths of example CA1 and PFC assemblies in relation to LFP events. Note the sequential organization of events with PFC leading CA1.

(E) Event probability fold change of spindles and ripples aligned to SO troughs (spindles, ***$p = 2.84 \times 10^{-14}$; PFC ripples, ***$p = 1.66 \times 10^{-7}$; CA1 ripples, ***$p = 3.86 \times 10^{-5}$, t test vs. 0), showing the spindle-PFC ripple-CA1 ripple timing.

(F) (Left) Cumulative proportion of event timings within ±250 ms of SO troughs (spindles vs. PFC ripples, ***$p = 2.41 \times 10^{-16}$, PFC ripples vs. CA1 ripples, ***$p = 4.62 \times 10^{-13}$), and (right) preferred SO phase for each event type (spindles vs. PFC ripples, $U^2 = 2.63$, ***$p = 5.73 \times 10^{-23}$, PFC ripples vs. CA1 ripples, $U^2 = 2.54$, ***$p = 3.61 \times 10^{-22}$, Watson's U^2 test). Gray shaded area indicates repeated data for visualization.

(G) SO-trough-aligned CA1 and PFC reactivation strength during SO-associated coordinated events where PFC ripples precede CA1 SWRs (PFC vs. shuffle, ***$p = 4.91 \times 10^{-19}$; CA1 vs. shuffle, ***$p = 2.12 \times 10^{-39}$; PFC vs. CA1 timing, ***$p = 7.96 \times 10^{-5}$).

(H) (Left) Relationship between independent PFC-ripple-associated CA1 suppression and SO up-state CA1 reactivation strength (data: $r = -0.40$, ***$p = 3.01 \times 10^{-21}$, shuffle: $r = -0.01$, $p = 0.86$), again showing a negative relationship. (Middle) SO-trough-aligned reactivation strength for CA1 assemblies. (Right) *Z* scored SO-trough-aligned CA1 reactivation strength plotted for the first and fourth quartiles.

See also Figure S6.

coordination affords the opportunity to study this widespread hippocampal phenomenon in greater detail and may be able to reconcile how SWR diversity may play a role in different aspects of cognition.[35–38]

The broad suppression that spans multiple cell types in CA1 during PFC ripples indicates that these cortical ripples have a unique contribution to the consolidation of recently acquired memories. The robust and predictive nature of this suppression suggests that PFC may have a role in modulating or biasing the content of hippocampal reactivation to influence what information to consolidate within the hippocampus and, in turn, the neocortex. Furthermore, modulation during PFC ripples may indicate a mechanism through which the E-I (excitation-inhibition) balance within the CA1 circuit is modified to maintain or restore a specific level of criticality[39,40] or to control the excitatory drive in the population of CA1 pyramidal cells,[41] thus increasing the signal to noise ratio for more robust information transfer.[42] Concurrently, CA1 suppression may also be a mechanism through which memory interference in the PFC is attenuated to allow for restricted cortical-ripple-mediated information transfer and plasticity throughout the neocortex to support memory integration through synchrony for semantic memories.[20,21,43] Indeed, previous studies have demonstrated that cortical areas are synchronized through ripple oscillations after learning and may be involved in binding neural activity across the cortex.[22,44] Notably, cortical high-frequency oscillations can be separated into slow and fast events and show preferential interaction with either hippocampal or cortical areas, respectively.[22] Given our finding of a relationship between CA1 suppression and PFC ripple frequency, effective consolidation may be dependent on a gradient of hippocampal-cortical communication mediated by high-frequency coupling. Similarly, spindle-uncoupled SWRs suppress activity in the mediodorsal thalamus, and the degree of suppression is correlated with SWR amplitude.[45] These results, along with our findings, suggest the presence of multiple suppressive circuit mechanism in the thalamocortical-hippocampal pathway for potentially modularly reducing interference. We hypothesize that PFC ripples may be associated with consolidation within cortical networks while suppressing hippocampal reactivation, and elucidation of the brain-wide processes associated with this suppression will require monitoring multiple cortical areas. However, it is important to note—given that the suppression of CA1 initiates prior to PFC reactivation during independent PFC ripple events—that there are likely other brain regions contributing to this process. Importantly, previous studies have shown that deep and superficial CA1 pyramidal neurons segregate into two molecularly defined populations that have distinct cortical targets and differentially engage SWRs and the theta oscillation.[37] The integration of our results thus raises the question of whether specific hippocampal-cortical subcircuits uniquely contribute to the dynamics surrounding the suppressive phenomenon that we report here. Future studies investigating this bidirectional modulation of CA1 in the context of the laminar distribution of cells (deep vs. superficial), their projection profiles to different cortical targets, such as the PFC and entorhinal cortex, and the state of plasticity (plastic vs. rigid) will yield a clearer picture of the multimodal dynamics present in the hippocampus.[37,46,47]

The finding that highly suppressed CA1 assemblies are strongly reactivated during coordinated ripples events, coupled with spindles and SOs—a salient signature of consolidation—also suggests a mechanism through which PFC prioritizes specific CA1 assemblies for consolidation based on future utility or generalizability through a cortical-hippocampal-directed process.[15,16,48,49] In contrast, previous studies have demonstrated a temporal interaction between the PFC and the hippocampus, where SWRs precede spindles to potentially tag neocortical traces for subsequent consolidation during spindles.[50] Although we did not find evidence for strong coupling between spindles and independent SWRs, other hippocampal subregions, such as the ventral CA1, may preferentially interact with cortical regions in a hippocampal-leading manner to support neocortical tagging.[51] Because SWRs in the dorsal and ventral CA1 are largely asynchronous during NREM sleep,[52] we predict that there exist sub-stages of memory consolidation mediated by bidirectional interactions between the PFC and distinct CA1 circuits. In the context of memory reconsolidation, suppression may be involved in preventing destabilization of certain salient CA1 memory traces.[53] Many studies of reconsolidation utilize experimental paradigms that are performed over the course of multiple days. It has been argued that specific changes in hippocampal dynamics potentially linked to consolidation, namely representational drift,[54] are a function of experience[55,56]; however, our longitudinal study design, wherein animals are exposed to multiple behavioral sessions in a day, can maintain a specific level of lability in CA1 that is consistent with circuit dynamics underlying reconsolidation. Given the persistent reactivation that occurs in CA1 subsequent to novel experience,[57] PFC-mediated suppression may be a mechanism through which CA1 activity is normalized to a level that is optimal for the induction of transcriptional processes that underlie memory maintenance or integration.[53]

Previous human fMRI studies have reported hippocampal suppression primarily during retrieval stopping, which is typically investigated in contexts where intrusive memories are proactively suppressed.[58,59] Our finding that coordinated-SWR-associated replay events have lower dimensionality and sequential structure raises the alternative possibility that the PFC may be involved in a process that prevents certain neural patterns from being consolidated through disruption of hippocampal sequential activity. SWRs are thought to facilitate retrieval through the reactivation of hippocampal ensembles, but the mechanisms through which these activity patterns are distinguished based on utility for consolidation are unclear. There must be a way for the brain to separate activity that is elicited by actual experience from novel, unrelated activity patterns that are a result of internally generated mechanisms to maintain a faithful rendering of reality.[8,60] Although these coordinated-SWR-associated replay events quantitatively represent sequences of behaviorally relevant activity, the up- and downstream brain areas conveying and receiving information, respectively, may play roles in determining the nature of the top-down influence of the PFC on hippocampal sequential activity.[34] This type of reality monitoring is crucial for the purported role of SWRs in imagination,[8] and further elucidation of the mechanisms underlying this process may lead to a deeper understanding of how it goes awry in neurological disorders such as schizophrenia.[60]

Overall, our results uncover a previously unknown PFC-ripple-mediated phenomenon with a distinct suppressive reactivation role that is suggestive of a modulatory process involved in modifying hippocampal representations and biasing hippocampal reactivation in sleep for effective future behavior. Given the procedural complexity of the integration of learned, novel experiences, the elucidation of the brain-wide neurophysiological phenomena that contribute to these processes at specific time points in different brain states and contexts is essential for understanding how interregional communication supports learning and may uncover a broader role for activity suppression in memory formation, consolidation, and retrieval.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- ○ Peri-SO CA1 and PFC reactivation strength
- QUANTIFICATION AND STATISTICAL ANALYSIS

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at https://doi.org/10.1016/j.cub.2024.05.018.

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AUTHOR CONTRIBUTIONS

J.D.S. and S.P.J. conceived the study. J.D.S. performed the experiments. J.D.S. processed and analyzed the data. J.D.S. wrote the initial manuscript. J.D.S. and S.P.J. reviewed and edited the manuscript. S.P.J. provided supervision and funding for all aspects of the study.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Chemicals		
Cresyl Violet	Acros Organics	Cat#: AC229630050
Formaldehyde	Fisher	Cat#: 50-00-0,67561,7732-18-5
Isoflurane	Patterson Veterinary	Cat#: 07-806-3204
Ketamine	Patterson Veterinary	Cat#: 07-803-6637
Xylazine	Patterson Veterinary	Cat#: 07-808-1947
Atropine	Patterson Veterinary	Cat#: 07-869-6061
Bupivacaine	Patterson Veterinary	Cat#: 07-890-4881
Euthasol	Patterson Veterinary	Cat#: 07-805-9296
Sucrose	Sigma-Aldrich	Cat#: S8501-5KG
Deposited Data		
Electrophysiology and behavior data	This paper	DANDI: 000978
Experimental Models: Organisms/Strains		
Rat: Long Evans	Charles River	Cat#: Crl:LE 006; RRID: RGD_2308852
Software and Algorithms		
MATLAB 2022a	Mathworks, MA	RRID: SCR_001622
Trodes	SpikeGadgets	http://www.spikegadgets.com
Matclust	Mattias P. Karlsson	https://www.mathworks.com/matlabcentral/fileexchange/39663-matclust, V1.7
Chronux	Partha Mitra	http://chronux.org/, V2.12
RobustICA[61]	Vicente Zarazoso, Pierre Comon	https://webusers.i3s.unice.fr/~zarzoso/robustica.html
Other		
Electrophysiology data acquisition system	SpikeGadgets	http://www.spikegadgets.com
12.7 μm NiCr tetrode wire	Sandvik	Cat#: PX000004

RESOURCES AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Shantanu P. Jadhav (shantanu@brandeis.edu).

Materials Availability

This study did not generate any new unique reagents.

Data and Code Availability

- Data underlying these results are available in the NWB (Neurodata Without Borders[62]) format on DANDI Archive (DANDI ID#: 000978).
- Custom MATLAB code used for analyses in this study is available on GitHub at (http://github.com/JadhavLab/PrefrontalRipples).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

All experimental procedures were approved by the Institutional Animal Care and Use Committee at Brandeis University and conformed to US National Institutes of Health guidelines. Twelve adult male Long-Evans rats (450-600 g, 4-6 months; RRID: RGD_2308852) were used in this study. Animals were individually housed and kept on a 12-hr regular light/dark cycle.

METHOD DETAILS

Behavior

Eleven out of the twelve animals were trained on a novel W-maze in a single day, as previously described.[63] Briefly, during the experimental day, all animals ran eight 15-20 minute behavioral sessions on a W-maze interleaved with nine 20-30 minute rest sessions in a sleep box. Reward (evaporated milk) was automatically delivered at each of the 3 reward wells upon completion of a correct trajectory. At the end of a behavioral session, animals were transferred to an opaque sleep box to rest.

Surgical implant and electrophysiology

Surgical implantation procedures were as previously described.[63] Animals (n = 12) were implanted with a microdrive array containing 30-64 independently moveable tetrodes targeting right dorsal hippocampal region CA1 (-3.6 mm AP and 2.2 mm ML) and right PFC (+3.0 mm AP and 0.7 mm ML) (Figures S1A and S1B). For 64 tetrode recordings (n = 1 animal), CA1 and PFC were targeted bilaterally. In one animal, ventral CA1 (-5.6 mm AP and 4.0 mm ML angled 12° mediolaterally) was targeted in addition to dorsal CA1 and PFC. On the days following surgery, hippocampal tetrodes were gradually advanced to the desired depths using characteristic EEG patterns (sharp wave polarity, theta modulation) and neural firing patterns as previously described.[63] A CA1 tetrode in corpus callosum served as hippocampal reference, and another tetrode in overlying cortical regions with no spiking signal served as prefrontal reference. A ground (GND) screw installed in the skull overlying cerebellum also served as a reference. All spiking activity and ripple-filtered LFPs (150-250 Hz; CA1 and PFC; see below) were recorded relative to the local reference tetrode. For detection of slower frequency components (i.e., theta, spindle, and slow oscillations), LFP was recorded with respect to the GND screw. Electrodes were not moved at least 4 hours before and during the recording day.

Data were collected using a SpikeGadgets data acquisition system (SpikeGadgets LLC). Spike data were sampled at 30 kHz and bandpass filtered between 600 Hz and 6 kHz. LFPs were sampled at 1.5 kHz and bandpass filtered between 0.1 Hz and 400 Hz. The animal's position and running speed were recorded with an overhead color CCD camera (30 fps) and tracked by color LEDs affixed to the headstage.

Note that only the 8 animals that successfully learned the task were used for analysis, with the exception of Figures S1J–S1M and S3H–S3K.

Histological verification of recording sites

At the completion of each experiment, electrolytic lesions were made by sending current (30uA) down 3 electrodes per tetrode for 3-4 seconds each. 24 hours later, animals were sacrificed by intraperitoneal injection of Euthasol and subsequent intracardial perfusion with 0.9% saline followed by 4% formaldehyde. The skull, with the attached drive, was submerged in 4% formaldehyde for 24 hours before tetrode retraction and brain extraction. Brains were stored in a 4% formaldehyde and 30% sucrose solution for at least 1 week before slicing with a freezing microtome (Leica) at 50uM. Slices were Nissl stained and imaged at 4x on a brightfield microscope (Keyence) and stitched together to generate composite images for lesion localization.

Spike sorting

Single units were sorted using a custom manual clustering program (MatClust, M. P. Karlsson) and were distinguished based on spike width, peak and trough amplitude, and principal components. Cluster quality was measured using isolation distance,[64] and only well isolated neurons were included for analysis.

Putative CA1 pyramidal cells (43.25 ± 3.06 pyramidal units per epoch) and interneurons were further separated by burst probability, mean firing rate, and trough-to-peak (Figures S2F–S2M). Burst probability was defined as the proportion of spikes within an epoch that occurred within 6 ms of each other.[65] All well isolated PFC units were included for analysis (34.36 ± 2.08 units per epoch).

Sleep state identification

To detect bouts of NREM and REM sleep, animals' head speed and CA1 LFPs were used as previously described.[19,66] During sleep box sessions, times of awake activity and immobility were defined as periods where head speed was greater or less than 4 cm/s, respectively. Candidate NREM sleep bouts were identified as periods where head speed remained <4 cm/s after an immobility period of at least 60 s. NREM and REM sleep were further identified and separated by using a theta to delta ratio metric. Briefly, CA1 LFP (referenced to GND) for each tetrode was bandpass filtered (8-12 Hz for theta; 1-4 Hz for delta) and averaged, and the ratio of theta to delta power was calculated. Periods that exceeded a set threshold (mean + 1 SD) for >10 s were categorized as REM bouts within candidate NREM bouts (Figure S1C).

LFP event detection

CA1 and PFC ripple detection: Ripple events were detected during immobility periods (<4 cm/s) as previously described.[19] Tetrodes with recorded cells in CA1 and PFC were used as electrodes for ripple detection. Each LFP from candidate electrodes were bandpass filtered in the ripple band (150-250 Hz), and the envelope power was calculated using a Hilbert transform and subsequently smoothed with a gaussian kernel (4 ms). Ripple events were detected as contiguous periods where the ripple power exceeded 3 SD of the mean on at least one tetrode. Events were further refined as periods around the detected event that remained above the mean (start and end times). The amplitude of each ripple event was reported as the number SDs above the mean.[12] To estimate the frequency of each ripple event, the power in each frequency bin was calculated around each ripple event (±500 ms) using multitaper time-frequency analysis (Chronux toolbox; http://chronux.org/) and was z-scored relative to a baseline spectrogram. The frequency of each event was defined as the frequency with the highest power in the ripple band (100-250 Hz) between the start and end time of each event.[67] Furthermore, ripple events in CA1 and PFC were designated as independent or coordinated based on whether cross-region event times (start to end intervals) overlapped. The coordinated events were further subset into leading or lagging events depending on the identity of the leading ripple. Only events where there was a clear temporal relationship were used (i.e. the start and end times of the leading event preceded the start and end times of the lagging event, respectively). Additionally, in PFC, isolated ripples were extracted as all events that were temporally isolated from both spindles and SOs by $\geq$500 ms.

PFC spindle detection: Spindle detection was performed using the same PFC tetrodes used for ripple detection. Similar to ripple detection, LFP from PFC tetrodes were bandpass filtered in the spindle band (12-16 Hz), and the envelope power was calculated using a Hilbert transform. Contiguous periods that exceeded 3 SD of the mean on at least one tetrode were considered spindle events. Events were further refined as periods around the detected event that remained above the mean (start and end times).

PFC slow wave detection: To extract slow waves from PFC LFP, tetrodes that were used to detect PFC ripple events were averaged together and filtered through two independent filters as previously described.[68] Briefly, the averaged LFP was filtered with a high pass Butterworth filter (0.1 Hz cutoff, 2nd order) followed by a low pass filter (4 Hz cutoff, 5th order). To separate SOs from other, lower amplitude slow waves (i.e. possibly delta oscillations), the following process was employed. Here, we are considering SOs as high amplitude waves in the 0.1-4 Hz range. First all zero crossings were identified and each segment that included an up state between two down states was extracted for further processing. For each epoch, a positive (>85 percentile of peaks) and negative threshold (<40 percentile of troughs) was set for down and up states, respectively. A wave was considered a SO if both the first peak was higher than the positive threshold and the trough was lower than the negative threshold. Slow waves were classified as low amplitude (other) if the trough was lower than the negative threshold but the first peak did not surpass the positive threshold. Furthermore, only events where the time from first peak to trough was 150-500 ms were included. To visualize the separation of SOs and other slow waves, K-Means clustering was performed on the mean peak and trough values from each epoch (Figure S6E).

All of the above events were further subset by periods of NREM sleep for subsequent analysis. For consistency, the same electrodes were used for detection of events across all epochs.

Ripple rates and ripple rate correlation

Ripple rates were calculated for each animal and epoch in sessions where the total duration of NREM sleep was >30 seconds. This yielded a rate vector for each animal, which was z-scored to account for the differences in ripple rate across animals.

In addition, to examine how changes in either CA1 or PFC ripple rate may drive changes in the occurrence of coordinated events in NREM sleep, the Pearson correlation was taken between the ripple rate vector calculated from all events and the rate vector for coordinated events. Within area ($CA1_{all}$-$CA1_{coord}$ or PFC_{all}-PFC_{coord}) r values were computed for each animal and compared.

To investigate changes in ripple rate within a sleep session (relative rate change), NREM sleep was split into two halves, 1st and 2nd, by dividing the total duration by 2. The ripple rates were calculated for each half, and the relative rate change was calculated as follows:

$$\%\ rate\ change\ =\ \left(\frac{rate_{2nd}\ -\ rate_{1st}}{rate_{1st}}\right)x100$$

Only epochs with a total NREM duration exceeding 120 seconds (60 s per rate calculation) were included for analysis.

Event cross-correlation

To examine whether independent and coordinated events are differentially coupled with spindle and SOs, the cross-correlation between event onset time vectors was calculated for each event pair (max time lag 4 seconds, 100 ms time bins) and smoothed with a Gaussian kernel (300 ms). For each comparison, ripple times were used as the reference vector. Thus, positive and negative lag values indicate that the compared LFP event tended to occur before and after ripple events, respectively.

To calculate 95% confidence intervals, both CA1 and PFC ripple times were jittered using the following steps. First, initial jitter values were resampled from the distribution of inter-event intervals separately for CA1 and PFC. Then, these values were randomly assigned negative or positive values to define the direction of the jitter. Finally, a value between 1 and 5 seconds was added to the direction of the jitter before calculating the cross-correlation. This process was repeated 1000 times for each epoch, and the 95th and 5th percentiles were calculated. This time jitter was chosen instead of a circular shuffle since only ripple times during NREM sleep were used for this calculation. A random circular shuffle would have likely shifted ripple times into periods not defined as NREM, yielding inaccurate results.

Ripple aligned modulation

Single unit ripple modulation in CA1 or PFC was calculated as previously described.[17] This analysis was restricted to neurons that fired >50 spikes cumulatively across all peri-ripple windows (±2 s). For each candidate neuron, an averaged ripple-triggered (CA1 or PFC ripple) time histogram (ripple-PSTH) was calculated. For comparison, 1000 shuffled ripple-PSTHs were generated by circularly shifting the spikes around each ripple event by a random amount. To determine whether a neuron was significantly modulated, the squared difference between the real ripple-PSTH and the shuffled data in the 0-200 ms window after ripple onset was calculated. In the case of CA1 modulation during independent PFC ripples, a larger window spanning ±200 ms around ripple onsets was used due to the observed CA1 population response aligned to PFC ripples (Figure S6D). If this real value exceeded 95% of the shuffled values, the neuron was categorized as significantly modulated. Furthermore, excited (EXC) and inhibited (INH) neurons were separated based on whether the modulation in ripple response window (either −200 to 200 or 0 to 200 ms) was positive (EXC) or negative (INH). Modulation index (MI) for each cell was calculated as the mean response in the background window (−600 to −200 ms prior to event onset) subtracted from the mean response in the response window.

To investigate the relationship between modulation of CA1 INH neurons and features of independent PFC ripple events (frequency, amplitude, and length), the following procedure was implemented. Ripple events in each epoch were split into frequency, amplitude, or length quartiles, and single unit ripple modulation was calculated as above. Then, the Pearson correlation coefficient was calculated between modulation index and quartile.

Spectral analysis

Wavelet spectral analyses were utilized to calculate the ripple aligned power spectra in PFC (2–350 Hz, Morlet wavelets). For each epoch, the PFC tetrode with the greatest number of detected PFC ripples was used. The power at each level of the wavelet transform was calculated around each ripple event (±500 ms) and was z-scored relative to a baseline spectrogram calculated for the entire epoch. For each animal, spectrograms from all sleep epochs were averaged (Figure S1D).

Ripple triggered spindle power

To determine whether there was a difference in spindle power during independent vs coordinated events, the envelope magnitude for each PFC spindle detection tetrode was z-scored and the maximum envelope power during each ripple event was calculated and averaged across electrodes.

Theta phase-locking

Theta phase-locking during active behavior on the W-Track was quantified using previously developed methods.[69] The CA1 reference tetrode located in corpus callosum was filtered in the 6-12 Hz range and was used to extract phase information, since theta phase has been reported to be consistent at different depths above the CA1 pyramidal layer.[70] Spikes were restricted to theta periods where the animals' speed was >5 cm/s, and spikes were assigned a phase between 0-360°. Only neurons with at least 100 spikes during theta periods were included in this analysis. The Rayleigh test was used to determine whether the phase distribution of each cell deviated from circular uniformity ($p < 0.05$ criterion). Subsequently, for each theta modulated cell, phase locking strength was measured using two separate methods, the circular concentration parameter, kappa (κ), and pairwise phase consistency (PPC).[71] κ was estimated by fitting the phase distributions to a von Mises distribution, and PPC was calculated by taking the dot product of the angular difference between each pair of phase values then averaging. To test for phase preference differences, the distributions were compared using a nonparametric Watson's U^2 test.

Single unit properties

Linearized spatial maps during active behavior on the W-Track were calculated during periods of high mobility (>5 cm/s; all SWR times excluded) at positional bins where occupancy was >20 ms.[17] Briefly, each rat's 2D position was projected onto a linearized skeleton of the W-Track and was classified as belonging to one of the four possible trajectories. Then, the linearized spatial maps were calculated by dividing the binned spike count by the occupancy in each 2 cm positional bin and smoothed with a Gaussian kernel (4 cm).

For each epoch, mean theta firing rate was calculated by dividing the number of spikes that occurred during mobility periods (>5 cm/s) by the total duration of mobility. Similarly, NREM firing rate was calculated by dividing the number of spikes during NREM (excluding SWR times) by the total duration.

To calculate the number of place fields for each CA1 neuron, 5,000 null linearized firing rate vectors were generated by randomly sampling each cell's spikes using the smoothed occupancy as the probability density function. The resulting spike vector (concatenated across all four trajectories) was smoothed with a Gaussian kernel (4 cm) and divided by the smoothed occupancy. Each bin that exceeded the 99th percentile of the null distribution was determined as having selective firing within that bin. To be considered a place field, at least 5 consecutive bins (10 cm) exceeding the 99th percentile was required.[46] Field width was calculated as the mean width across all place fields for each cell.

Spatial information, I (bits/spike), was calculated using the following formula:

$$I_{\text{spike}} = \sum_{i=1}^{N} p_i \frac{\lambda_i}{\lambda} \log_2 \frac{\lambda_i}{\lambda}$$

where $i = 1, \ldots, N$ represents the spatial bin, p is the occupancy probability of bin i, and λ is the mean firing rate of the neuron.

To calculate the field stability of modulated CA1 neurons, the Pearson correlation of the concatenated linearized firing fields was taken between the two W-Track epochs flanking the sleep epoch during which the cells were modulated.

Single unit SWR activity

The firing rate gain of each neuron was calculated by dividing the firing rate during CA1 SWR events by the mean rate across all NREM bouts in an epoch. Additionally, to exclude the contribution of SWR firing to the baseline NREM firing rate, the mean rate was calculated by excluding spiking activity across all SWR events during NREM sleep. To determine SWR burst probability for each cell, the number of SWRs where burst spikes were detected (at least 2 spikes <6 ms apart) was divided by the total number of SWR events in which that cell was active. Similarly, SWR spike latency was estimated by taking the average latency to first spike during each SWR event the cell was active. For quartile analysis, cells were sorted by modulation index and split into 4 equal groups. The corresponding SWR spiking metrics were then averaged and plotted for each group. To determine whether there was a relationship between the modulation indices and burst probability or spike latency, the Pearson correlation coefficient was calculated.

Event aligned multi-unit activity

To characterize population activity aligned to LFP events, spike trains of each neuron were binned into 5 ms time windows. Activity was aligned to PFC spindles, SOs, ripples, and CA1 ripples. The event aligned multi-unit activity was then normalized by the mean population firing rate during NREM sleep.

For vCA1 recordings, multiunit activity was extracted as all events >70 μV on vCA1 tetrodes.

Ripple coactivity

To quantify ripple coactivity during independent and coordinated events, a pairwise coactivity z-score was calculated as previously described.[72] Briefly, this coactivity measure estimates how likely two cells will fire together during ripple events and is normalized by the incidence of spiking independent of the other cell:

$$Zcoactivity_{a,b} = \frac{n_{ab} - \frac{n_a n_b}{N}}{\sqrt{n_a n_b (N - n_a)(N - n_b) / \left(N^2 (N - 1)\right)}}$$

where N is the number of ripple events, n_a and n_b are the number of events where cell a and cell b were active, respectively, and n_{ab} is the number of events where both cells were active. Epochs where there were >20 independent and coordinated events were included in the analysis.

Assembly detection

To detect cell assemblies in CA1 and PFC populations, a previously described method was used.[73,74] Significant co-firing patterns in CA1 or PFC were detected using a method based on principal and independent component analyses. For each W-Track epoch, the spike trains of each neuron were binned into 20 ms time windows (excluding CA1 ripple times) and Z-scored to eliminate any biases due to differences in firing rates. Principal component analysis was then applied to the Z-scored spike matrix (Z). The resulting eigenvalue decomposition of the correlation matrix (C) was given by:

$$\sum_{i=1}^{n} \lambda_i p_i p_i^T = C$$

Where p_i is the i^{th} principal component of C and λ_i is the corresponding eigenvalue. Note that $C = \frac{1}{n} ZZ^T$ is the correlation matrix of Z. To determine the number of significant coactivity patterns in the data, the Marcenko-Pastur law was used to calculate a threshold eigenvalue λ_{max}, which was given by $\lambda_{max} = \left(1+\sqrt{n/B}\right)^2$, where n is the number of recorded units and B is the total number of bins. Note that an eigenvalue exceeding λ_{max} indicates that the corresponding cell assembly explains more correlation than expected if the activity of neurons was independent of each other. The number of eigenvalues above λ_{max}, N_A, represents the number of significant patterns found in the data. These principal components were then projected back onto the original z-scored spike data:

$$Z_{PROJ} = P_{SIGN}^T Z$$

where P_{SIGN} is the nXN_A matrix with N_A columns. Independent component analysis (ICA, RobustICA)[61] was then used to identify patterns that were maximally independent from each other. ICA was applied to the matrix Z_{PROJ} to find an $N_A X N_A$ unmixing matrix, W, which was used to obtain the weights of each cell in each assembly.

$$V = P_{SIGN} W$$

Since the sign of the output of ICA is arbitrary, the signs of the weight vector were set such that the highest absolute weight was set to positive. Cells with weights greater than the mean + 2SD were considered member cells. Furthermore, field similarity for member vs non-member cell pairs was calculated by taking the Pearson's correlation between the concatenated linearized firing fields (Figure S4A).

In addition to the assembly weights, a contribution metric, I, was calculated to assess the relative influence of EXC and INH cells on assembly reactivation during sleep. To quantify the contribution of a cell, c, the reactivation strength of each assembly, k, was recalculated after omitting the spike train of that cell, R_k^{-c}, to yield two separate reactivation strength vectors, R_k^{-c} and R_k. The contribution was then calculated by:

$$I_k^c = \frac{1}{2}\left(1 - \frac{\overline{R_k^{-c}}}{\overline{R_k}}\right)$$

where ½ is a normalization factor such that the sum of the contributions equals 1.[74]

Assembly reactivation strength

To assess the reactivation of these detected assemblies during sleep sessions, the expression strength of each pattern over time was given by:

$$R_k(t) = z(t)^T P_k z(t)$$

$R_k(t)$ is the reactivation strength of assembly k at time t, $z(t)$ is the z-scored firing rate vector for all neurons at time t, and P_k is the projection matrix for assembly k, constructed by taking the outer product of its weight vector v_k. Additionally, the diagonal of the projection matrix was set to zero to reduce the influence of highly active neurons. Thus, only patterns of coactivity are detected as periods of high reactivation strength. For CA1-PFC assembly analysis, within region correlations in the projection matrices were also set to 0 to measure only the coactivity between areas. Also, only CA1-PFC assemblies that had at least one member cell in each area were included for analysis. Only sleep sessions with a preceding run session to use as a template were included for reactivation analysis (sleep sessions 2-9).

Assembly coactivation rate

To investigate CA1-PFC assembly coactivity during independent and coordinated CA1 ripple events, a coactivation rate was calculated by first defining discreet periods of high reactivation ($R > 5$). For each ripple event, the number of CA1 and PFC assemblies that were active was calculated separately for each area. The events were then subset to include events where both CA1 and PFC assemblies were active. The minimum number of active assemblies across areas was used as the number of coactivation events (i.e., CA1-2, PFC-3 would be 2 coactivation events). The total number of events across all ripples was divided by the cumulative ripple time to obtain a coactivation rate per epoch.

Ripple triggered assembly strength

The reactivation strength of each assembly aligned to ripple onsets was determined by calculating the mean in a ±8 s window across all ripple triggered events. The mean was then z-scored, and the average assembly strength within the peak bins (±200 ms) was calculated to obtain a single reactivation strength value for each assembly. Using this method to calculate overall reactivation strength yielded a gradient of strengths as expected (Figures S4I and S4J). For epoch grouped analysis, epochs 2-4, 5-6, and 7-9 were used for early, middle, and late sleep sessions, respectively.

Assembly reinstatement

To assess assembly reinstatement from one W-track session to another, the above assembly reactivation analysis was performed, but instead, cell identity was matched across three consecutive epochs (run-sleep-run) to measure the same assembly over time. The epochs were matched such that each W-track session acted as the template epoch, except for the 8th W-track session. Reinstatement was defined as the difference in the average expression strength during the two compared exposures (2nd exposure minus 1st exposure).[73]

Replay analysis

Bayesian decoding for replay analysis was performed as previously described.[75] Only sleep sessions with a preceding run session to use as a template were included for replay analysis (sleep sessions 2-9). A memoryless (uniform prior probability) Bayesian decoder was built for each of the four trajectory types to estimate the probability of animals' position given the observed spikes (Bayesian reconstruction; or posterior probability matrix):

$$P(X, tr|spikes) = \frac{P(spikes|X, tr)P(X, tr)}{P(spikes)}$$

where X is the set of all linear positions on the track for different trajectory types ($tr \in$ (center-to-right, right-to-center, center-to-left, left-to-center)), and we assumed a uniform prior probability over X and tr. Assuming that all N place cells active during a candidate event fired independently and followed a Poisson process:

$$P(spikes|X, tr) = \Pi_{i=1}^{N} P(spikes_i|X, tr) = \Pi_{i=1}^{N} \frac{(\tau f_i(X, tr))^{spikes_i} e^{-\tau f_i(X, tr)}}{spikes_i!}$$

where τ is the duration of the time window (15 ms), $f_i(X,tr)$ is the expected firing rate of the ith cell as a function of sampled location X and trial type tr, and $spikes_i$ is the number of spikes of the ith cell in a given time window. Therefore, the posterior probability matrix can be derived as follows:

$$P(X,tr|spikes) = C\left(\Pi_{i=1}^{N} f_i(X,tr)^{spikes_i}\right)e^{-\tau\sum_{i=1}^{N} f_i(X,tr)}$$

where C is a normalization constant.

We identified replay during SWRs based on the Bayesian decoder described above. First, candidate events were defined as the SWR events longer than 50 ms during which $\geq$ 5 place cells fired. Each candidate event was then divided into 15-ms non-overlapping bins, and the posterior probability matrix was computed based on the Bayes' rule. To determine replay significance, we employed the two following shuffle-based methods.

Replay significance (Monte Carlo method)

In this first method, the p-value was calculated based on the Monte Carlo shuffle.[76] First, we drew 10,000 random samples from the posterior probability matrix for each decoded bin and assigned the sampled locations to that bin. Linear regression was then performed on the bin number versus the location points. The resulting R-squared was compared with 1,500 regressions, in which the time bins were shuffled). A candidate event with $p < 0.05$ based on the time shuffle was considered as a replay event. Since the shuffling procedure measured how well the decoded positions along SWR time matched a behavioral trajectory, we considered the trajectory with the lowest p-value from the shuffling procedure as the replay trajectory, and its R-squared was reported as replay quality (Figure S5F).

Replay Significance (Weighted Correlation vs. shuffle)

Additionally, a sequence score was computed for each candidate replay event as the correlation between position and time, weighted by the posterior probability.[46,77] A weighted mean was calculated for each event by:

$$m(pos;Pr) = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} Pr_{ij}pos_j}{\sum_{i=1}^{M}\sum_{j=1}^{N} Pr_{ij}}$$

The weighted covariance by:

$$cov(pos,bin;Pr) = \frac{\sum_{i=1}^{M}\sum_{j=1}^{N} Pr_{ij}(pos_j - m(pos|Pr))(bin_i - m(bin|Pr))}{\sum_{i=1}^{M}\sum_{j=1}^{N} Pr_{ij}}$$

Where pos_j is the jth spatial bin, bin_i is the ith time bin, M is the number of time bins, N is the number of spatial bins, and Pr_{ij} is the posterior probability for the spatial bin in that time bin.

Finally, the weighted correlation was given by:

$$r(pos,bin;Pr) = \frac{cov(pos,bin|Pr)}{\sqrt{cov(pos,pos|Pr)(cov(bin,bin|Pr))}}$$

Subsequently, 1,000 shuffled spatial template matrices were constructed by circularly shifting each cell's template independently. The entire Bayesian decoding process was performed using these shuffled templates to yield a null distribution of weighted correlation values for each candidate replay event. An event with $p < 0.05$ based on the spatial template shuffle was considered as a significant replay event. Additionally, for each event, a sequence score (rZ) was calculated by:

$$rZ = \frac{|r(observed)| - \overline{|r(null)|}}{S.D.(|r(null)|)}$$

Where S.D. is the standard deviation of the null distribution. Note that the rZ value does not signify directionality of the replay event.

Jump distance for each significant event was reported as the distance between the maximum decoded probability of two adjacent time bins. Each pair of time bins yielded a distance value unless there were bins with no spiking activity. The maximum of these values was taken and normalized by dividing by the length of the corresponding trajectory.

Per cell contribution (PCC) analysis

PCC was calculated for each cell to assess the influence of a cell's participation in a replay event.[46] For each candidate replay event, each cell's place representation was circularly shuffled separately such that only one cell's spatial representation was shuffled at a time. Each cell's representation was shuffled 1000 times to calculate an $rZ_{cellshuffle}$ value as above. Then, PCC was given by:

$$PCC_{e,c} = [(rZ_e(observed) - rZ_{e,c}(cellshuffle_c)]x\#Participants_e$$

Where '$\#Participants_e$' is the number of cells active during event e. The average PCC was calculated for cells that were active in >10 significant replay events per epoch. For the calculation of PCC difference between coordinated and independent events, the cell was required to be active in >5 events of each type per epoch.

Rank-order correlation

To evaluate the consistency with which populations of neurons in CA1 fired in sequence, we calculated the rank-order correlation between each event and its template as previously described.[24,78] For each SWR event with >5 cells active, the spiking activity (both pyramidal cells and interneurons included) was reduced to each unit's first spike time, chronologically ordered, and normalized from 0 to 1. Then, a cross-validated template-based method was used to calculate sequence similarity between each event and its template. For each event, a template was created by averaging the normalized rank of each cell across all events excluding the event in question. The rank correlation was reported as the Pearson correlation coefficient between the event and template. Templates were constructed separately for independent and coordinated events, utilizing only events of one type.

Population dimensionality during SWRs

To investigate the dimensionality of population activity during independent and coordinated SWR events, principal component analysis (PCA) was performed on separate spiking activity matrices. Sessions with >20 units (pyramidal and interneurons) were included for analysis. Since independent events outnumbered coordinated events, we only included sessions where there were >100 events of each type. Then, we randomly resampled each activity matrix 100 times (50 samples at a time) and applied PCA to each matrix to obtain the variance explained by each component (n components = n cells). We considered the population dimensionality to be the proportion of components required to explain >90% of the variance.

Ripple type prediction

Ripple type (independent or coordinated) in CA1 and PFC was predicted by fitting a linear model (MATLAB, fitclinear) to the areas' spike count data. Since the independent events largely outnumbered the coordinated events, the event counts were equalized prior to training by randomly excluding a subset of independent events. The binary decoders were built using 10-fold cross validation, and prediction accuracy was assessed by comparing performance against models constructed with randomly permuted spike matrices (n = 1000). To account for the dropped events, this entire process was repeated 10 times for each animal and epoch. This method was also used to predict different coordinated events (CA1 or PFC leads), but instead, both CA1 and PFC spiking activity was used.

Replay probability during SOs

To calculate the probability of CA1 replay events during the down or up state of SOs, the occurrence of replay start times of each significant event (independent, coordinated, or all) was determined and divided by the total number of replay events in NREM sleep for each epoch.

Peri-SO event probability

To investigate the sequential occurrence of LFP events during NREM sleep, the fold change occurrence around SO troughs was calculated for PFC spindles, ripples, and CA1 ripples. Event start times were separately binned in 20 ms time bins in a response window ±2 seconds (*resp*) centered around SO troughs (up states). The event probability in each bin was then calculated, smoothed with a 60 ms Gaussian kernel, and compared to the mean probability across all bins in the background window −3 to −2 seconds (*bck*) around SO trough. The fold change in each bin was then calculated by:

$$foldchange = \frac{(P_{resp} - \overline{P_{bck}})}{\overline{P_{bck}}}$$

Since this approach results in a single fold change vector, we utilized a resampling method to determine whether the resultant peaks were not due to chance. The occurrence matrix was randomly sampled (100 samples at a time), and the probability fold change was calculated as above with the corresponding background bins. This process was repeated 50 times to obtain a distribution of values in each time bin. Then, the bin in which the peak fold change occurred was compared to 0 with a t-test.

To compare the timing of events aligned to SO troughs, the distributions of all event times within ±250 ms of SO troughs were compared. To further validate this timing result, distributions of the times of peak fold change occurrences around SO troughs were compared by calculating event fold change separately for each epoch and animal.

To investigate the SO phase preference of spindles and ripples, phase information for each discreet SO event (upstate and downstate) was extracted. Each LFP event (event onset) that occurred within the start/end interval of SOs was assigned a phase. Then, a circular von Mises probability density function was computed based on parameters (preferred direction and concentration parameter) estimated from the distribution of phase angles. To test for phase preference differences for spindles and ripples, the distributions were compared with Watson's U2 test.

Peri-SO CA1 and PFC reactivation strength

To investigate the temporal dynamics of CA1 and PFC reactivation during these events, we looked at CA1 lagging events surrounding SO troughs. Reactivation strength of each assembly was aligned to SO troughs where there was a CA1 lagging ripple event within ±250 ms of a SO trough. As a comparison, each assembly was circularly shuffled, and the peak reactivation bin of the real data was compared to the same bin of the shuffled data. In addition, to compare the relative reactivation times of CA1 and PFC assemblies, the peak of the mean reactivation strength for each was taken in the 0-200 ms time window surrounding SO troughs, and the timing distributions were compared. The same process was performed for CA1 leading events, but in this case, timing differences were assessed using a ±100 ms window around SO troughs. The windows were determined by examining the reactivation profiles surrounding SO troughs.

To compare CA1 assembly suppression during independent PFC ripples to CA1 reactivation during SO-associated coordinated events where CA1 ripples lag, the suppression strength of each assembly in a ±200 ms window around PFC ripples was calculated as in "*Ripple triggered assembly strength*." Then, the mean reactivation strength in a 0-200 ms window around SO troughs was compared to the suppression during PFC ripples for each assembly.

QUANTIFICATION AND STATISTICAL ANALYSIS

All statistical analyses were performed with MATLAB (MathWorks, Natick, MA; R2022a) functions or custom scripts. No specific analysis was performed to predetermine sample size, but the number used in this study is similar to or larger than typically used. Nonparametric, two-tailed statistical tests (Wilcoxon rank-sum and signed-rank, for unpaired and paired data, respectively) were used throughout the paper unless noted. $P < 0.05$ was considered the cutoff for statistical significance. Boxplots show the median (center line), interquartile range (box), and the furthest points not considered outliers (whiskers). Although not displayed in some cases, outliers were included in all statistical analyses. Unless otherwise noted, values and errors in the text denote means ± SEM. p-values in the text are reported as follows: $p > 0.05$, *$p < 0.05$, **$p < 0.01$, ***$p < 0.001$.